

BAYESIAN ESTIMATION OF THE PROCESS CAPABILITY INDEX C_{pk}

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Summary: In this paper theoretical and simulation results are derived for the posterior distribution of the process capability index C_{pk} under two different but related prior distributions. For the conventional prior, the exact posterior moments of the index are calculated. Using these moments, Pearson curve and Cornish-Fisher approximations to the posterior distribution are obtained for a real problem. Gibbs sampling is used in the case of the reference (probability matching) prior to obtain the unconditional posterior distribution of C_{pk} . Finally, it is shown that the probability matching prior for the distribution function (a function of the index) is the same as the probability matching prior for the index itself.

1. Introduction

Process capability indices have been widely used in the manufacturing industry. They measure the ability of a manufacturing process to produce items that meet certain specifications. Application examples include the manufacturing of semiconductor products (Hoskins et al., 1988), jet-turbine engine components (Hubele et al., 1991), wood products (Lyth and Rabiej, 1995), audio speaker drivers (Chen and Pearn, 1997) and many others.

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There is a need to understand and interpret process capability indices. In the literature on statistical quality control there have been some attempts to study the inferential aspects of these indices, see Kane (1986), Chan et al. (1988), Chan and Owen (1989), Zhang et al. (1990) and Pearn et al. (1992). Most of the existing work in this area has been devoted to classical frequentist large sample theory.

From a Bayesian point of view, Cheng and Spiring (1989) have derived the posterior distribution of the capability index $C_p = \frac{\ell_1 - \ell_0}{6\sigma}$ under normality which corresponds to a conventional "non-informative" prior. C_p relates the allowable spread of the process $\ell_1 - \ell_0$ to its natural spread; customarily taken to be 6σ . According to Bernardo and Irony (1996) a similar Bayesian analysis for the $C_{pk} = \min\left(\frac{\ell_1 - \mu}{3\sigma}, \frac{\mu - \ell_0}{3\sigma}\right)$ index has not been done (ℓ_0 = lower specification limit, ℓ_1 = upper specification limit, μ = process mean and σ = process standard deviation). Bernardo and Irony (1996), on the other hand, proposed a general multivariate Bayesian process capability index that includes the conventional index as a limiting case. This measure focuses on the proportion of future parts expected to lie outside the tolerance limits rather than on a single summary value that describes the combined effect of these characteristics. Niverthi and Dey (2000) proposed a vector valued process capability index with each component separately describing individual characteristics and used the Bayesian paradigm to model the data.

2. Definitions and notation

Two of the commonly used process capability indices are $C_p = \frac{\ell_1 - \ell_0}{6\sigma}$ and $C_{pk} = \min\left(\frac{\ell_1 - \mu}{3\sigma}, \frac{\mu - \ell_0}{3\sigma}\right)$. According to Bernardo and Irony (1996)

C_{pk} is the normalized distance between the process mean and its closest specification limit. It can easily be verified that $C_{pk} = C_p(1 - w)$ where $w = \frac{2|m - \mu|}{\ell_1 - \ell_0}$ and $m = \frac{\ell_1 + \ell_0}{2}$ is the midpoint of the specification limits. Thus, C_{pk} modifies C_p by a standardized measure w of non-centrality of the process and $C_{pk} = C_p$ if and only if the process is centred at m .

The larger the value of C_{pk} , the more capable is the process. In general, if the value of a process capability index is greater than 1.0 the process is said to be capable. According to Niverthi and Dey (2000), the thrust these days in the manufacturing industry is to achieve a C_{pk} value of at least 1.33. The definition of C_{pk} includes as special cases those processes for which only one limit exists, by letting either $\ell_0 \rightarrow -\infty$ or $\ell_1 \rightarrow \infty$, in which case it reduces to the appropriate standardized measure.

In the next section we will derive the exact posterior moments (mean, variance, third and fourth central moments) for the process capability index $C_{pk} = \frac{\mu - \ell_0}{3\sigma}$ (which is applicable when there is no upper limit). By applying these moments and Pearson curves or Cornish-Fisher expansions, it will be shown in Section 4 that approximations of the exact posterior distribution of C_{pk} can be obtained.

3. Exact posterior moments of C_{pk}

Let $y_1, \dots, y_n$ be an independent sample from a manufacturing process. In this paper it will be assumed that the y_i ($i = 1, \dots, n$) are independently, identically normally distributed with mean μ and variance σ^2 . Since both μ and σ^2 are unknown and no prior information is available, the conventional 'non-informative' default prior

$$\pi(\mu, \sigma^2) \propto \sigma^{-2} \quad (3.1)$$

will be specified for them in this section. Using (3.1), it is well known (see for example Zellner, 1971) that the conditional posterior density of μ is normal:

$$\mu | \sigma^2, \mathbf{y} \sim N\left(\bar{y}, \frac{\sigma^2}{n}\right) \quad (3.2)$$

and in the case of the variance component σ^2 , the posterior density is given by

$$p(\sigma^2 | \mathbf{y}) = K (\sigma^2)^{-\frac{1}{2}(n+1)} \exp\left\{-\frac{1}{2}(n-1)s^2/\sigma^2\right\} \sigma^2 > 0, \quad (3.3)$$

an Inverted Gamma density, where $\mathbf{y} = [y_1 y_2 \dots y_n]'$, $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ is the sample mean, $s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$ is the sample variance and

$$K = \left\{ \frac{(n-1)s^2}{2} \right\}^{\frac{1}{2}(n-1)} \frac{1}{\Gamma\left(\frac{n-1}{2}\right)} \quad (3.4)$$

is a normalizing constant. The following theorem can now be stated.

THEOREM 3.1

The mean, variance, third and fourth central moments of the posterior distribution of the process capability index $C_{pk} = \frac{\mu - \ell_0}{3\sigma}$ are given by

$$m'_1 = \left(\frac{\bar{y} - \ell_0}{3}\right) \frac{\Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{n-1}{2}\right)} \left\{ \frac{2}{(n-1)s^2} \right\}^{\frac{1}{2}} \quad (3.5)$$

$$m_2 = \frac{1}{9n} + \frac{2(\bar{y} - \ell_0)^2}{9(n-1)s^2} \left\{ \frac{(n-1)}{2} - \frac{\Gamma^2\left(\frac{n}{2}\right)}{\Gamma^2\left(\frac{n-2}{2}\right)} \right\} \quad (3.6)$$

$$m_3 = \frac{(\bar{y} - \ell_0)^3}{27} \left\{ \frac{2}{(n-1)s^2} \right\}^{3/2} \frac{\Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{n-1}{2}\right)} \left\{ \frac{2\Gamma^2\left(\frac{n}{2}\right)}{\Gamma^2\left(\frac{n-1}{2}\right)} - \frac{2n-3}{2} \right\} \quad (3.7)$$

and

$$\begin{aligned} m_4 = & \frac{1}{27n^2} + \frac{(\bar{y} - \ell_0)^2}{27n} \frac{2}{(n-1)s^2} \left\{ (n-1) - \frac{\Gamma^2\left(\frac{n}{2}\right)}{\Gamma^2\left(\frac{n-1}{2}\right)} \right\} \\ & + \frac{(\bar{y} - \ell_0)^4}{27(n-1)^2(s^2)^2} \left[\frac{1}{3}(n-1)(n+1) + \frac{4\Gamma^2\left(\frac{n}{2}\right)}{\Gamma^2\left(\frac{n-1}{2}\right)} \right. \\ & \left. \left\{ \frac{(n-3)}{3} - \frac{\Gamma^2\left(\frac{n}{2}\right)}{\Gamma^2\left(\frac{n-1}{2}\right)} \right\} \right] \quad (3.8) \end{aligned}$$

Proof: The proof is given in the Appendix.

4. Approximated posterior distributions of the capability index C_{pk}

In this section Pearson curve and Cornish-Fisher approximations of the posterior distribution of C_{pk} will be derived. For details on how to determine the parameters of a Pearson curve given the values of its moments see, for example, Elderton (1953) or Elderton and Johnson (1969). A type I Pearson curve is used to approximate the posterior distribution of C_{pk} for our data set.

The Cornish-Fisher percentage points can be calculated as follows:

The standardized version of the capability index C_{pk} is $T = \frac{C_{pk} - m'_1}{\sqrt{m'_2}}$ and the level α percentage point of T is defined as t_α . With this definition

it follows that a Cornish-Fisher expansion (see Cornish and Fisher, 1937 and Fisher and Cornish, 1960) for t_α is given by

$$\tilde{t}_\alpha = z_\alpha + \frac{1}{6}\ell_3(z_\alpha^2 - 1) + \frac{1}{24}\ell_4(z_\alpha^3 - 3z_\alpha) - \frac{1}{36}\ell_3^2(2z_\alpha^3 - 5z_\alpha)$$

where z_α is the corresponding percentage point of the standard normal distribution, $\ell_r = \frac{\kappa_r}{(\kappa_2)^{r/2}}$ ($r = 3, 4$) and κ_r is the r th cumulant of C_{pk} . Also

$$\kappa_2 = m_2, \quad \kappa_3 = m_3 \quad \text{and} \quad \kappa_4 = m_4 - 3m_2^2$$

where m'_1, m_2, m_3 and m_4 are defined in equations (3.5)–(3.8).

The percentage point of level α of C_{pk} is then given by $t_\alpha\sqrt{m_2} + m'_1$.

5. Monte Carlo simulation procedure for estimating the posterior distribution of C_{pk}

From the definition of the capability index and using the fact that $\mu \mid \sigma^2, \mathbf{y} \sim N\left(\bar{y}, \frac{\sigma^2}{n}\right)$ (equation (3.2)), it follows that

$$C_{pk} \mid \sigma^2, \mathbf{y} \sim N\left\{\frac{\bar{y} - \ell_0}{3\sigma}, \frac{1}{9n}\right\}. \quad (5.1)$$

From (3.3) it follows that

$$p(\sigma^2 \mid \mathbf{y}) = K(\sigma^2)^{-\frac{1}{2}(n+1)} \exp\left\{-\frac{1}{2}(n-1)s^2/\sigma^2\right\}$$

which is an Inverted Gamma distribution with normalizing constant K defined in (3.4). Standard routines are used in the simulation procedure:

- (i) Simulate τ from a χ_{n-1}^2 distribution, as a sum of $(n-1)$ squared independent normal random deviates and calculate $\sigma^2 = \frac{(n-1)s^2}{\tau}$.
- (ii) Given σ^2 , calculate the conditional posterior density function

$p(C_{pk} | \sigma^2, y)$ which is defined in (5.1).

- (iii) Repeat steps (i) and (ii) $\tilde{\ell}$ times. For our example $\tilde{\ell}$ was taken as 10 000. Using a Rao-Blackwell argument (see Gelfand and Smith, 1991), a density estimate of the unconditional posterior distribution of $C_{pk} = \frac{\mu - \ell_0}{3\sigma}$ can be obtained by averaging $p(C_{pk} | \sigma^2, y)$ over the $\tilde{\ell}$ repetitions.

6. Probability matching and reference priors for

C_{pk}

The Bayesian paradigm is attractive in many types of statistical problems – especially so in capability index problems, but the choice of an appropriate non-informative prior distribution has been controversial. Common non-informative priors in multiparameter problems, such as Jeffreys' prior, can have features that have an unexpectedly dramatic effect on the posterior. Recently Datta and Ghosh (1995) derived the differential equation that a prior must satisfy if the posterior probability of a one-sided credibility interval for a parametric function and its frequentist probability are to agree up to $O(n^{-1})$ where n is the sample size. They proved that the agreement holds if and only if

$$\sum_{\alpha=1}^m \frac{\partial}{\partial \theta_{\alpha}} \{ \eta_{\alpha}(\theta) \pi(\theta) \} = 0 \quad (6.1)$$

where $\pi(\theta)$ is the *probability matching prior* for θ , the vector of unknown parameters, and where

$$\eta(\theta) = \frac{I^{-1}(\theta) \Delta_t(\theta)}{\sqrt{\nabla_t'(\theta) I^{-1}(\theta) \nabla_t(\theta)}} = [\eta_1(\theta), \dots, \eta_m(\theta)]' \quad (6.2)$$

with

$$\nabla_t = \left[\frac{\partial}{\partial \theta_1} t(\theta), \dots, \frac{\partial}{\partial \theta_m} t(\theta) \right]' \quad (6.3)$$

It is clear that $\eta'(\theta) I(\theta) \eta(\theta) = 1$ for all θ where $I^{-1}(\theta)$ is the inverse of $I(\theta)$, the Fisher information matrix of θ per unit observation, and $t(\theta)$ is the parameter of interest. The following theorem can now be stated.

THEOREM 6.1

Let $\{y_i, i = 1, \dots, n\}$ be independently normally distributed with parameter vector $\theta = [\mu, \sigma]^T$. Suppose the parameter of interest is the process capability index $C_{pk} = \frac{\mu - \ell_0}{3\sigma} = t(\theta)$. Then the probability matching prior for C_{pk} is given by

$$\pi^M(\mu, \sigma^2) = \sigma^{-3} \left\{ 1 + \frac{(\mu - \ell_0)^2}{2\sigma^2} \right\}^{-\frac{1}{2}} \quad (6.4)$$

Proof: The proof is given in the Appendix.

The Jeffreys' prior is not always suitable for multiparameter problems. In recognition of this problem Berger and Bernardo (1992), proposed the *reference prior* approach to the development of non-informative priors, the key feature of which was a possible dependence of the reference prior on specification of parameters of interest and nuisance parameters. In this section the reference prior of Berger and Bernardo (1992) will be derived for the process capability index. The solution depends on the ordering of the parameters and how the parameter vector is partitioned into sub-vectors. In spite of these difficulties, there is growing evidence, mainly through examples, that reference priors provide "sensible" answers from a Bayesian point of view and some more limited evidence that frequentist properties of inference from reference posteriors are asymptotically "reasonable". We will also examine whether the reference priors satisfy the probability-matching criterion (6.1).

The following theorem can now be stated.

THEOREM 6.2

For the process capability index $C_{pk} = \frac{\mu - \ell_0}{3\sigma}$, the reference prior relative to the ordered parameterization (μ, σ^2) is given by

$$\pi^R(\mu, \sigma^2) = \sigma^{-3} \left\{ 1 + \frac{(\mu - \ell_0)^2}{2\sigma^2} \right\}^{-\frac{1}{2}}. \quad (6.5)$$

Proof: The proof is given in the Appendix.

COROLLARY

The reference prior is also a probability matching prior. This follows from equation (6.4). Mukerjee and Dey (1993), on the other hand, derived priors ensuring up to $O(n^{-1})$ frequentist validity of the posterior quantiles of the parameter of interest. Their proofs, especially those for the frequentist calculations, are much simpler than those of previous researchers. If the reference prior and the probability matching prior produce the same results, they suggested higher order asymptotics to obtain the appropriate prior.

7. Gibbs sampling

For the reference (probability matching) prior of the capability index, the joint posterior density of μ and σ^2 is given by

$$p(\mu, \sigma^2 | \mathbf{y}) \propto (\sigma^2)^{-\frac{1}{2}(n-1)} e^{-\frac{1}{2}(n-1)s^2/\sigma^2} \left(\frac{n}{\sigma^2}\right)^{\frac{1}{2}} \frac{e^{-\frac{n}{2}(\bar{y}-\mu)^2/\sigma^2}}{\sqrt{2\pi}} \times \\ \sigma^{-3} \left\{ 1 + \frac{(\mu - \ell_0)^2}{2\sigma^2} \right\}^{-\frac{1}{2}}. \quad (7.1)$$

Technical difficulties arising in the calculation of the marginal posterior densities needed for Bayesian inference in the case of reference and probability matching priors have long served as an impediment to the wider application

of the Bayesian framework to data analysis. The reason for this is that the integration operation plays a fundamental role in Bayesian statistics.

Recently, due to work of Gelfand and Smith (1991), Gelfand et al. (1990), Carlin et al. (1992) and Gelfand et al. (1992), the Gibbs sampler has been shown to be a useful tool for applied Bayesian inference in a broad variety of statistical problems. The typical objective of the sampler is to collect a sufficiently large number of parameter realizations from conditional posterior densities in order to obtain accurate estimates of the marginal posterior densities.

In the $(t(\theta), \sigma^2)$ parameterization the conditional densities are given by

$$p(t(\theta) | \sigma^2, \mathbf{y}) \propto \left\{ 1 + \frac{9}{2} t^2(\theta) \right\}^{-\frac{1}{2}} e^{-\frac{n}{2} (\bar{y} - 3\sigma t(\theta) - \ell_0)^2 / \sigma^2} -\infty < t(\theta) < \infty \quad (7.2)$$

and

$$p(\sigma^2 | t(\theta)) \propto (\sigma^2)^{-\frac{1}{2}(n+2)} e^{-\frac{n}{2} (\bar{y} - 3\sigma t(\theta) - \ell_0)^2 / \sigma^2} \quad \sigma^2 > 0. \quad (7.3)$$

The Gibbs sampler can now be implemented. Random numbers are generated using rejection sampling. The process is repeated $\tilde{\ell}$ times. For the problems in this paper we used $k = 30$, $\tilde{\ell} = 10\,000$ and $k = 20$, $\tilde{\ell} = 20\,000$. Using the Rao-Blackwell argument mentioned in Section 5, a density estimate of the unconditional posterior distribution of $t(\theta) = C_{pk} = \frac{\mu - \ell_0}{3\sigma}$ can be obtained.

8. Probability matching prior for $\delta = \Phi\left(\frac{\mu - \ell_0}{\sigma}\right)$

A parameter that is often of interest in capability index theory is $\delta = \Phi\left(\frac{\mu - \ell_0}{\sigma}\right)$. Bernardo and Irony (1996) used the conventional non-informative prior $\pi(\mu, \sigma^2) \propto \sigma^{-2}$, but mentioned on page 15 of their article

that this prior is not necessarily the most appropriate “default” prior for δ . In this section we will therefore derive the probability matching prior for δ . The following theorem can now be stated.

THEOREM 8.1

Let $\{y_i, i = 1, \dots, n\}$ be independently normally distributed with parameter vector $\theta = [\mu, \sigma]$. Suppose the parameter of interest is $\delta = \Phi\left(\frac{\mu - \ell_0}{\sigma}\right)$ where Φ is the standard normal distribution function. Then the probability matching prior for δ is given by

$$\pi_{\delta}^M(\mu, \sigma^2) = \sigma^{-3} \left\{ 1 + \frac{\mu - \ell_0}{2\sigma^2} \right\}^{-\frac{1}{2}} \quad (8.1)$$

Proof: The proof is given in the Appendix.

9. An application

To illustrate the calculation of the process capability index C_{pk} using the Bayesian techniques discussed above, we consider the following aircraft data collected by the Pratt and Whitney company. These data are given in the first column of Table 1 on page 672 of Niverthi and Dey (2000). The unit of measurement is centimetres. Since the component is critical, the manufacturing process and subsequent steps must have a high degree of precision ($\ell_0 = 6.393$ and $\sigma_0^2 = 10^{-7}$).

9.1 Results

The non-informative prior was used in the Pearson’s curve and Rao-Blackwell methods (Monte Carlo simulation). For the reference prior the Gibbs sampling method was used. Figures 1 and 2 illustrate the posterior distributions of C_{pk} for the data set given in Table 1. The graphs look the same because of the large set of data used. The histogram results are also shown in Figure 1. It

is further clear that the Pearson curve approximation is quite good. Under the Jeffreys' prior, the mean C_{pk} is 2.54. A 95% credibility interval for C_{pk} is (2.04, 3.06). Under the reference (probability matching) prior, the mean C_{pk} is 2.63. A 95% credibility interval for C_{pk} is (2.11, 3.11).

Figures 3 and 4 illustrate the posterior distribution of C_{pk} for the first 20 observations given in Table 1. Because of the smaller number of observations used, the posterior distributions in Figures 3 and 4 have a larger variance when compared to the corresponding Figures 1 and 2. The Jeffreys' prior gives rise to a mean C_{pk} of 2.94 with a 95% credibility interval for C_{pk} of (2.02, 3.94). The reference (probability matching) prior gives rise to a mean C_{pk} of 3.15 and a 95% credibility interval for C_{pk} of (2.21, 4.11).

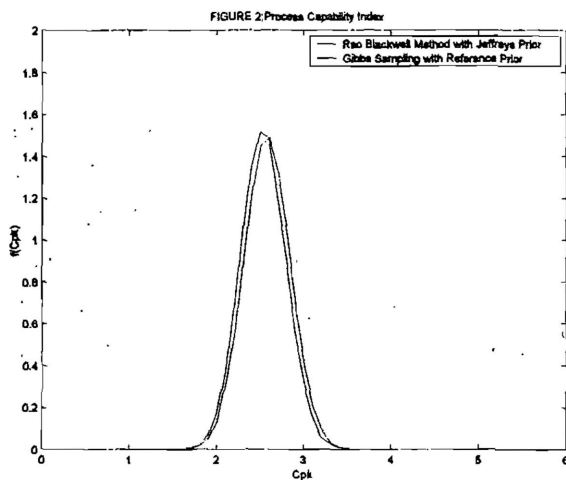
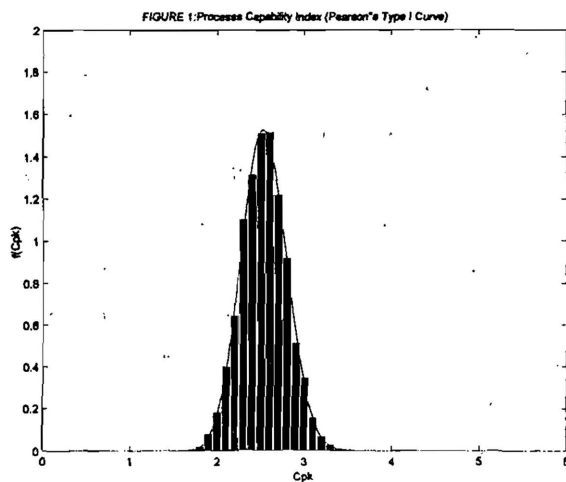
From the results it is clear that the different priors give different estimates. We recommend the probability matching (reference) prior because it is designed to produce posterior credible intervals which are asymptotically identical to their frequentist counterparts.

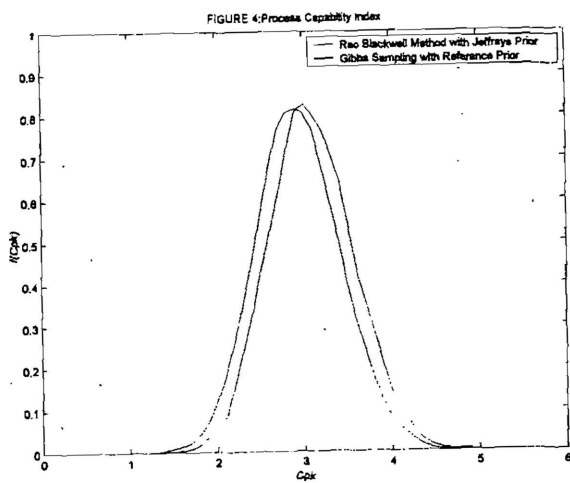
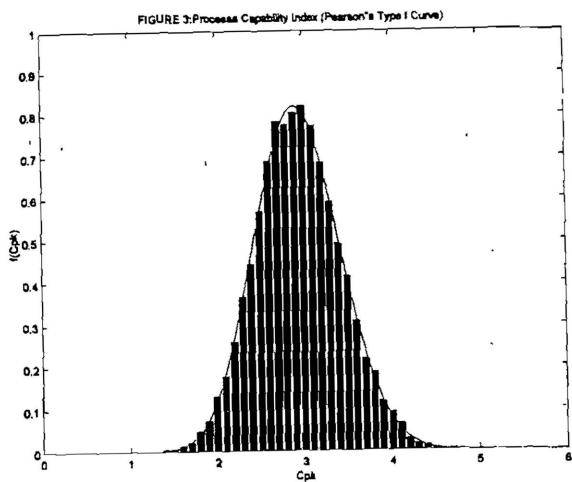
10. Conclusion

Bayesian inference has a number of attractive features. A full Bayesian analysis provides a natural way of taking into account all sources of uncertainty in the estimation of parameters. In this paper uncertainty about the true value of the process capability index is incorporated into the analysis through the choice of a non-informative prior distribution. The probability matching prior is recommended because it is designed to produce posterior credible intervals that are asymptotically identical to their frequentist counterparts.

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Appendix

Proof of Theorem 3.1

Since $\mu \mid \sigma^2, y \sim N\left(\bar{y}, \frac{\sigma^2}{n}\right)$. The posterior density of $C_{pk} = \frac{\mu - \ell_0}{3\sigma}$ conditionally on σ^2 is normal with mean $\frac{\bar{y} - \ell_0}{3\sigma}$ and variance $\frac{1}{9n}$. Denote the first four posterior moments about the origin for C_{pk} (conditionally on σ^2) by μ_1, μ_2', μ_2'' and μ_4' and the central moments by μ_2, μ_3 and μ_4 . Then

$$\mu_1' = \frac{\bar{y} - \ell_0}{3\sigma} \quad (\text{A.1})$$

$$\mu_2' = \frac{1}{9} \left\{ \frac{1}{n} + \frac{(\bar{y} - \ell_0)^2}{\sigma^2} \right\} \quad (\text{A.2})$$

$$\mu_3' = \frac{(\bar{y} - \ell_0)}{27} \left\{ \frac{3}{n\sigma} + \frac{(\bar{y} - \ell_0)^2}{\sigma^3} \right\} \quad (\text{A.3})$$

and

$$\mu_4' = \frac{1}{27} \left[\frac{1}{n^2} + (\bar{y} - \ell_0)^2 \left\{ \frac{2}{n\sigma^2} + \frac{(\bar{y} - \ell_0)^2}{3\sigma^4} \right\} \right]. \quad (\text{A.4})$$

Also from (3.3) and (3.4) it follows that the r th posterior moment of σ^{-2} about the origin is given by

$$E \left\{ (\sigma^{-2})^r \mid y \right\} = K \left\{ \frac{2}{(n-1)s^2} \right\}^{\frac{1}{2}(n+r-1)} \Gamma \left(\frac{n+r-1}{2} \right),$$

where K is defined in equation (3.4). By substituting for $r = 1, 2, 3$ and 4 in equations (A.1) to (A.4) and using the relationships between moments about the origin and central moments, equations (3.5) to (3.8) follow.

Proof of Theorem 6.1

The Fisher Information matrix of $\theta = [\mu, \sigma]'$ per unit observation is

$$I(\mu, \sigma) = I(\theta) = \text{diag} [\sigma^{-2}, 2\sigma^{-2}] \text{ and } I^{-1}(\mu, \sigma) = \text{diag} \left[\sigma^2, \frac{1}{2}\sigma^2 \right].$$

Consider

$$C_{pk} = t(\theta) = \frac{\mu - \ell_0}{3\sigma}. \quad \text{Then } \frac{\partial t(\theta)}{\partial \mu} = \frac{1}{3\sigma} \quad \text{and} \quad \frac{\partial t(\theta)}{\partial \sigma} = -\frac{(\mu - \ell_0)}{3} \frac{1}{\sigma^2}.$$

Therefore,

$$\nabla'_t(\theta) = \left[\frac{1}{3\sigma} \frac{\ell_0 - \mu}{3\sigma^2} \right] = \frac{1}{3\sigma} \left[1 \frac{\ell_0 - \mu}{\sigma} \right].$$

Now

$$\nabla'_t(\theta) I^{-1}(\theta) = \frac{\sigma}{3} \left[1 \frac{\ell_0 - \mu}{2\sigma} \right]$$

and

$$\sqrt{\nabla'_t(\theta) I^{-1}(\theta) \nabla_t(\theta)} = \frac{1}{3} \sqrt{1 + \frac{(\ell_0 - \mu)^2}{2\sigma^2}},$$

which means that

$$\eta'(\theta) = \frac{\nabla'_t(\theta) I^{-1}(\theta)}{\sqrt{\nabla'_t(\theta) I^{-1}(\theta) \nabla_t(\theta)}} = [\eta_1(\theta) \quad \eta_2(\theta)]$$

where

$$\eta_1(\theta) = \frac{\sigma}{\sqrt{1 + \frac{(\ell_0 - \mu)^2}{2\sigma^2}}} \quad \text{and} \quad \eta_2(\theta) = \frac{\ell_0 - \mu}{2\sqrt{1 + \frac{(\ell_0 - \mu)^2}{2\sigma^2}}}.$$

The prior $\pi(\theta) = \pi^M(\mu, \sigma) = \left\{ 1 + \frac{(\ell_0 - \mu)^2}{2\sigma^2} \right\}^{\frac{1}{2}} \sigma^{-2}$ is a probability-matching prior because it satisfies the differential equation (6.1).

Proof of Theorem 6.2

$$C_{pk} = \frac{\mu - \ell_0}{3\sigma} = t(\theta).$$

Define

$$A = \frac{\partial(\mu, \sigma)}{\partial(t(\theta), \sigma)} = \begin{bmatrix} 3\sigma & 3t(\theta) \\ 0 & 1 \end{bmatrix}.$$

Hence the Fisher Information matrix under the reparameterization $(t(\theta), \sigma)$ is given by

$$I(t(\theta), \sigma) = A' I(\mu, \sigma) A = \begin{bmatrix} 9 & 9\sigma^{-1}t(\theta) \\ 9\sigma^{-1}t(\theta) & 9t^2(\theta)\sigma^{-2} + 2\sigma^{-2} \end{bmatrix}.$$

Following the notations in Berger and Bernardo (1992) and Bernardo (1998), the functions h_j ($j = 1, 2$) which are needed to calculate the reference prior for the group ordering $(t(\theta), \sigma)$ can be obtained from $I(t(\theta), \sigma)$ as follows:

$$h_1^{\frac{1}{2}} = \left| 9 - 9\sigma^{-1}t(\theta) (9t^2(\theta)\sigma^{-2} + 2\sigma^{-2})^{-1} 9\sigma^{-1}t(\theta) \right|^{\frac{1}{2}} \\ (18)^{\frac{1}{2}} \{9t^2(\theta) + 2\}^{-\frac{1}{2}}$$

and

$$h_2^{\frac{1}{2}} = \sigma^{-1} \{9t^2(\theta) + 2\}^{\frac{1}{2}}.$$

Therefore, the reference prior relative to the ordered parameterization $(t(\theta), \sigma)$ is given by

$$\pi^R(t(\theta), \sigma) \propto \left\{ 1 + \frac{9}{2}t^2(\theta) \right\}^{-\frac{1}{2}} \sigma^{-1}.$$

In the (μ, σ) parameterization this corresponds to

$$\pi^R(\mu, \sigma) \propto \left\{ 1 + \frac{1}{2} \frac{(\mu - \ell_0)^2}{\sigma^2} \right\}^{-\frac{1}{2}} \sigma^{-2}$$

which proves the theorem. See also Bernardo (1998) pages 60–61 for a similar proof.

Proof of Theorem 8.1

$$\delta = \Phi\left(\frac{\mu - \ell_0}{\sigma}\right) = \int_{-\infty}^{\frac{\mu - \ell_0}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz.$$

Let

$$\frac{\mu - \ell_0}{\sigma} = \nu. \text{ Then } \frac{\partial \Phi(\nu)}{\partial \mu} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\nu^2} \left(\frac{1}{\sigma}\right)$$

and

$$\frac{\partial \Phi(\nu)}{\partial \sigma} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\nu^2} \left(-\frac{\ell_0 - \mu}{\sigma^2}\right).$$

Define

$$f = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\nu^2}, \text{ then } \nabla'_t(\theta) = f \begin{bmatrix} 1 & \frac{\ell_0 - \mu}{\sigma} \end{bmatrix}$$

Since

$I^{-1}(\theta) = I^{-1}(\mu, \sigma) = \text{diag} [\sigma^2, \frac{1}{2}\sigma^2]$, it follows that

$$\nabla'_t(\theta) I^{-1}(\theta) = \sigma^2 f \left[1 \quad \frac{\ell_0 - \mu}{2\sigma} \right]$$

and

$$\sqrt{\nabla'_t(\theta) I^{-1}(\theta) \nabla_t(\theta)} = \sigma f \left[1 + \frac{1}{2} \left(\frac{\mu - \ell_0}{\sigma} \right)^2 \right]^{\frac{1}{2}}$$

which means that

$$\eta_1(\theta) = \frac{\sigma}{\sqrt{1 + \frac{1}{2} \left(\frac{\mu - \ell_0}{\sigma} \right)^2}} \quad \text{and} \quad \eta_2(\theta) = \frac{-\frac{1}{2}(\mu - \ell_0)}{\sqrt{1 + \frac{1}{2} \left(\frac{\mu - \ell_0}{\sigma} \right)^2}}.$$

Since $\eta_1(\theta)$ and $\eta_2(\theta)$ are exactly the same as the corresponding functions in Theorem 6.1, the probability matching priors for $C_{pk} = \frac{\mu - \ell_0}{3\sigma}$ and

$\delta = \Phi \left(\frac{\mu - \ell_0}{\sigma} \right)$ are similar and are given by

$$\pi^M(\mu, \sigma^2) = \left\{ 1 + \frac{(\mu - \ell_0)^2}{2\sigma^2} \right\} \sigma^{-3}$$

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